Frozen Core Isolation and Quantum-Resistant Cryptographic Commitments from Planted k-SAT

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Abstract

We prove that for planted k-SAT instances with $k \geq 7$ at clause density $\alpha/\alpha_s \geq 0.21$, a positive fraction of variables are frozen directly in the planted model—without requiring transfer from the random model via quiet planting. The expected number of "support clauses" per variable (clauses in which that variable is the unique satisfying literal) exceeds 1 at remarkably low density: $\alpha/\alpha_s \approx 0.20$ for $k = 7$, compared to the random-model freezing threshold at $\alpha_f/\alpha_s \approx 0.90$. We prove that the resulting frozen-core structure implies topological disconnection of the solution subgraph across cluster boundaries, with a cycle-robustness argument showing that short cycles in the factor graph cannot quench the supercritical repair cascade. As an immediate corollary, the Hilbert space spanned by satisfying assignments decomposes into orthogonal sectors preserved by any unitary generated by the adjacency matrix—blocking quantum walks, QAOA at all depths, and quantum annealing. We construct a post-quantum commitment scheme whose binding property reduces to the hardness of solving planted k-SAT, provide formal proofs of completeness, soundness, and zero-knowledge, and derive a digital signature scheme with existential unforgeability via the Fiat–Shamir transform. We present a six-vector quantum attack analysis with proved barriers against five algorithmic families. We give concrete parameter recommendations at NIST security levels 1, 3, and 5, and position the scheme within the landscape of SAT-based and CSP-based cryptographic constructions. We prove that the Grover query complexity for breaking the binding property is $\Omega(2^{fn/2})$; empirical cryptanalysis of Glucose and MiniSat CDCL solvers on our exact distribution yields a classical attack cost of $2^{0.234n}$ operations, enabling concrete parameter selection at NIST security levels 1, 3, and 5. Empirical validation across 100 random seeds at $n = 16$ confirms complete cluster isolation at every instance tested.

1 Introduction

Post-quantum cryptography rests on a narrow foundation. The three hardness families underlying deployed schemes—lattices, hash functions, and error-correcting codes—represent a concentration of risk: a breakthrough against any one family would compromise a large fraction of the post-quantum infrastructure. Diversifying the set of available hardness sources is a security imperative [5]. This concern is sharpened by recent work proposing additional hardness families—quantum stabilizer decoding [19] and, in the present paper, frozen-core constraint satisfaction—suggesting the viable assumption space is broader than the current standards reflect.

We introduce a fourth hardness family based on the cluster structure of random constraint satisfaction problems. Our main contributions are:

Theorem 1 (Frozen Core Isolation)**.** *For planted k-SAT with $k \geq 7$ and clause density α satisfying $e_{\sup}(k, \alpha) > 1$ (equivalently, $\alpha/\alpha_s > 1/\big(k \cdot 2^k \ln 2 \cdot (2^k - 1)^{-1}\big) \approx 0.20$ for $k = 7$), the following hold with high probability:*

(i) Each variable has expected support count $e_{\sup} = k\alpha/(2^k - 1) > 1$, where a support clause is one in which the variable is the unique satisfying literal under the planted solution y^*.

(ii) A fraction $f = 1 - e^{-e_{\sup}(1-q)} > 0$ of variables are frozen, where $q < 1$ is the extinction probability of a Galton–Watson process with offspring mean $e_{\sup}$.

(iii) The connected component $\mathrm{Comp}(y^*)$ of y^* in the Hamming-1 solution subgraph G_{SAT} is contained in $B(y^*, (1 - f + \varepsilon)n)$ for any $\varepsilon > 0$.

(iv) For two planted solutions y_1, y_2 with $\mathrm{Ham}(y_1, y_2) > 2(1 - f)n$, the connected components $\mathrm{Comp}(y_1)$ and $\mathrm{Comp}(y_2)$ are disjoint.

Corollary 2 (Quantum Isolation)**.** *Under the hypotheses of Theorem 1, the Hilbert space $\mathcal{H} = \mathrm{span}\{|s\rangle : s \text{ satisfies } \Phi\}$ decomposes as $\mathcal{H} = \bigoplus_i \mathcal{H}_i$, where each $\mathcal{H}_i$ corresponds to a connected component of G_{SAT}. Any unitary of the form $g(A)$, where A is the adjacency matrix of G_{SAT} and g is any analytic function (including e^{-iAt}), preserves this decomposition.*

Theorem 3 (Commitment and Signature Schemes)**.** *There exists a post-quantum commitment scheme with perfect completeness, soundness error $2^{-\lambda}$ (for security parameter λ), and computational zero-knowledge, whose binding property reduces to the hardness of solving planted k-SAT. The underlying sigma protocol yields, via the Fiat–Shamir transform, a digital signature scheme with existential unforgeability under chosen-message attack (EUF-CMA) in the random oracle model.*

The paper does not claim quantum circuit lower bounds for planted k-SAT. No existing post-quantum scheme has such a proof. What we provide is, in certain respects, more explicit quantum resistance evidence than is available for existing schemes: proved barriers against five of six major quantum algorithmic families (Table 5), with only generic brute-force search (Grover) remaining as an unstructured threat mitigated by standard parameter doubling.

1.1 Related work and positioning

Goldreich's one-way function. Goldreich [17] proposed planted constraint satisfaction problems as candidates for one-way functions. His construction evaluates a fixed d-ary predicate over an expander graph; security evidence comes from exponential lower bounds against myopic backtracking algorithms [8] and statistical algorithm lower bounds [13]. Our construction differs in three ways: (a) we use standard planted k-SAT rather than a custom predicate on an expander, (b) our hardness mechanism is topological—frozen-core cluster isolation—rather than algorithmic-class-specific, and (c) we provide explicit quantum resistance analysis (Table 5) with no counterpart in the Goldreich OWF literature.

Feldman–Perkins–Vempala. Feldman, Perkins, and Vempala [13] proved tight $\Omega(n^{r/2})$ lower bounds for statistical algorithms on planted k-SAT, matching known upper bounds. Their results provide evidence for the hardness of *finding* the planted solution. Our Theorem 1 addresses a complementary question: even if an adversary finds *some* satisfying assignment, the frozen-core structure guarantees it lies in the same cluster as the planted solution, making the binding property of our commitment scheme structurally enforced. The FPV statistical algorithm bounds strengthen our hardness assumption by ruling out a broad class of classical attacks.

SAT-based cryptography. Schmittner [24] proposed a public-key encryption scheme using a random SAT formula as public key and a satisfying assignment as private key. That construction lacks any structural hardness analysis—no planted model, no freezing threshold computation, no cluster isolation, and no quantum security argument. More recently, Cho et al. [7] proposed "Equilibrium SAT Based PQC" using a counting/multiset mechanism over SAT instances. Neither work exploits the phase-transition structure of random CSPs; both treat SAT as a generic NP-hard problem. Our approach is fundamentally different: hardness derives not from the generic NP-completeness of SAT but from the specific geometric structure (frozen clusters, topological disconnection) that emerges at controlled clause densities in the planted model.

Frozen variables and hardness. The connection between frozen variables and computational hardness was conjectured by Zdeborová and Krzakala [26] and studied rigorously for random CSPs by Achlioptas and Ricci-Tersenghi [2], Molloy [20], and Coja-Oghlan and Zdeborová [9]. All prior analysis concerns the *random* model. Our Lemma 6 provides, to our knowledge, the first computation of the freezing threshold in the *planted* model, revealing that planted instances freeze at roughly 1/4.4 the density required in the random model (for $k = 7$). Munoz-Bauza et al. [23] explicitly identified the formal connection between frozen variables and algorithmic hardness as an open problem in the context of planted Ising instances; our Theorem 1 resolves this for planted k-SAT.

New PQC hardness families. Lu, Poremba, Quek, and Ramkumar [19] recently proposed quantum stabilizer decoding as a new post-quantum hardness family, arguing it is plausibly incomparable to LPN. Our work is parallel in motivation—diversifying the PQC hardness landscape beyond lattices, hashes, and codes—but based on entirely different mathematics. Together these efforts suggest that the space of viable PQC assumptions is broader than the current NIST portfolio reflects.

2 Preliminaries

2.1 Planted k-SAT

Fix integers n (variables) and k (clause width). A planted k-SAT instance is generated as follows:

1. Choose a planted solution $y^* \in \{0,1\}^n$ uniformly at random.
2. Generate $m = \lfloor \alpha n \rfloor$ clauses independently: each clause selects k distinct variables uniformly and k literal signs uniformly, conditioned on y^* satisfying the clause.

The satisfiability threshold is $\alpha_s \approx 2^k \ln 2$ [11].

2.2 Cluster structure and frozen variables

For α above the clustering threshold α_d, the set of satisfying assignments decomposes into exponentially many well-separated clusters [11, 1]. Above the freezing threshold $\alpha_f > \alpha_d$, each cluster contains a positive fraction of frozen variables [2, 9].

Definition 4 (Frozen variable)**.** *Variable x_i is frozen in cluster C if x_i takes the same value in every solution reachable from C's core solution by a path of single-flip satisfying assignments. Equivalently, x_i is constant on the connected component of C's core in G_{SAT}.*

Definition 5 (Solution subgraph)**.** *$G_{\mathrm{SAT}} = (S, E)$ has vertex set $S = \{s \in \{0,1\}^n : s$ satisfies $\Phi\}$ and edge set $E = \{\{s,t\} : \mathrm{Ham}(s,t) = 1\}$.*

3 Proof of Theorem 1

The proof proceeds in five steps: (1) compute the support count directly in the planted model, (2) establish tree-likeness of the planted factor graph, (3) prove that supercritical support implies freezing via Warning Propagation, (4) derive Hamming ball containment, and (5) conclude component disjointness.

3.1 Step 1: Support count in the planted model

This is the key technical contribution. We analyze the planted clause distribution directly, without transferring results from the random model.

Lemma 6 (Planted support count)**.** *In a planted k-SAT instance at density α, the expected number of support clauses for a variable x_i (clauses in which x_i is the unique satisfying literal under y^*) is*

$$e_{\mathrm{sup}} = \frac{k\alpha}{2^k - 1}.$$

Proof. Consider a clause c containing variable x_i. The number of clauses containing x_i is $\mathrm{Bin}(m, k/n)$, with expectation $k\alpha$.

For such a clause, the literal for x_i has sign $+1$ or -1 with equal probability before conditioning on acceptance (i.e., y^* satisfying the clause). Let p_{acc} denote the acceptance probability.

If x_i appears with sign matching y_i^* ("positive"): the clause is always accepted, since x_i provides one satisfying literal. Probability of this sign: $1/2$. Conditional acceptance: 1.

If x_i appears with sign opposite to y_i^* ("negative"): the clause is accepted only if at least one of the other $k-1$ literals is satisfied by y^*. Probability: $1/2$. Conditional acceptance: $1 - 2^{-(k-1)}$.

Thus $p_{\mathrm{acc}} = \frac{1}{2} \cdot 1 + \frac{1}{2}(1 - 2^{-(k-1)}) = 1 - 2^{-k}$.

Given that c is accepted and contains x_i, the probability that x_i is the unique satisfying literal is:

$$P(\text{unique}) = \frac{P(x_i \text{ positive} \mid \text{acc}) \cdot P(\text{all other } k{-}1 \text{ lits false} \mid x_i \text{ pos})}{1} = \frac{1/2}{1 - 2^{-k}} \cdot 2^{-(k-1)} = \frac{1}{2^k - 1}.$$

Multiplying by the expected number of clauses containing x_i:

$$e_{\mathrm{sup}} = k\alpha \cdot \frac{1}{2^k - 1}.$$

□

Remark 7 (Planted vs. random freezing threshold)**.** *Setting $e_{\mathrm{sup}} = 1$ gives the planted freezing threshold $\alpha_f^{\mathrm{plant}} = (2^k - 1)/(k \cdot 2^k \ln 2)$. For $k = 7$: $\alpha_f^{\mathrm{plant}}/\alpha_s \approx 0.204$. The random-model freezing threshold is $\alpha_f^{\mathrm{rand}}/\alpha_s \approx 0.898$ [2, 9]. The planted model freezes at 4.4 times lower density than the random model. At our operating point $\alpha/\alpha_s = 0.78$: $e_{\mathrm{sup}} \approx 3.81$, deeply supercritical.*

Table 1: Planted freezing threshold $\alpha_f^{\text{plant}}/\alpha_s$ vs. random-model estimate $\alpha_f^{\text{rand}}/\alpha_s$.

k	e_{sup} formula	$\alpha_f^{\text{plant}}/\alpha_s$	$\alpha_f^{\text{rand}}/\alpha_s$	Ratio
5	$5\alpha/31$	0.280	0.719	0.39
7	$7\alpha/127$	0.205	0.898	0.23
9	$9\alpha/511$	0.160	0.967	0.17
15	$15\alpha/32767$	0.096	0.999	0.10

3.2 Step 2: Tree-likeness of the planted factor graph

Lemma 8 (Planted tree-likeness)**.** *The factor graph of planted k-SAT at density $\alpha < \alpha_s$ has the same vertex-edge topology as a uniformly random k-uniform hypergraph with $m = \alpha n$ hyperedges on n vertices. In particular, the depth-r neighborhood of any variable node is a tree with probability $1 - o(1)$ for $r = o\big(\sqrt{\log n/\log(k\alpha)}\big)$.*

Proof. In the planted model, each clause selects k variables uniformly at random and then selects literal signs conditioned on y^* satisfying the clause. The conditioning affects signs only, not which variables appear in which clause. The hypergraph structure (clause-variable incidence) is therefore identically distributed to the standard random k-uniform hypergraph. The result follows from the standard local tree-likeness theorem for random hypergraphs [10, 21]. □

Remark 9. *At the densities relevant to our construction ($k\alpha \approx 484$ for $k = 7$, $\alpha/\alpha_s = 0.78$), the tree-like depth is $r = O(1)$. This suffices for Warning Propagation, which converges in $O(1)$ iterations when the clause-to-variable ratio exceeds the freezing threshold [9].*

3.3 Step 3: Supercritical support implies freezing

Lemma 10 (Frozen–disconnected)**.** *For planted k-SAT at density α with $e_{\text{sup}} > 1$, let F denote the set of variables receiving at least one hard warning under Warning Propagation on the factor graph. Then w.h.p.:*

(a) $|F| \geq fn$ for $f = 1 - e^{-e_{\text{sup}}(1-q)} > 0$, where $q < 1$ is the extinction probability of a Galton–Watson process with offspring distribution $\text{Poi}(e_{\text{sup}})$.

(b) For each $x_i \in F$, the value $x_i = y_i^*$ is constant on $\text{Comp}(y^*)$.

Proof of (a). On the tree-like factor graph (Lemma 8), Warning Propagation has a unique fixed point. A variable x_i receives a hard warning from clause c if x_i is the unique satisfying literal in c under y^*. By Lemma 6, the number of hard warnings per variable converges to $\text{Poi}(e_{\text{sup}})$ on the local tree. The fraction of variables receiving ≥ 1 hard warning is $1 - e^{-e_{\text{sup}}} > 0$.

Not every hard warning produces a frozen variable; x_i is frozen only if the repair cascade triggered by flipping x_i diverges. The cascade is a Galton–Watson process with offspring mean e_{sup}: each flipped variable breaks its own support clauses, each of which requires repairing a different variable. For $e_{\text{sup}} > 1$ the process is supercritical, with survival probability $1 - q > 0$. The fraction of variables for which at least one warning triggers a surviving cascade is $f = 1 - e^{-e_{\text{sup}}(1-q)}$.

Concentration follows by McDiarmid's inequality: the frozen status of each x_i depends on the $O(1)$ clauses in its depth-1 neighborhood, and changing a single clause affects at most k variables'

frozen status. Setting $F = |\{i : x_i \text{ frozen}\}|$ as a function of m independent clause choices, each with bounded difference k:

$$\Pr\big[|F - fn| > n^{2/3}\big] \le 2\exp\Big(-\frac{2n^{4/3}}{mk^2}\Big) = 2\exp\big(-\Omega(n^{1/3})\big) \to 0.$$

Therefore $|F| = fn \pm o(n)$ w.h.p. □

Proof of (b). Suppose for contradiction that $s \in \mathrm{Comp}(y^*)$ satisfies $s_i \neq y_i^*$ for some $x_i \in F$. Then there exists a path $y^* = s_0, s_1, \ldots, s_T = s$ in G_{SAT}. Let t^* be the first index at which x_i is flipped. At step t^*, all support clauses for x_i become violated. Each must be repaired by flipping another variable in a subsequent step, but each repair flip breaks that variable's own support clauses.

First-passage property. Any repair sequence flips each variable at most once: a second flip of x_j would re-break the clause whose violation caused the first flip of x_j. Therefore the cascade defines a tree T of distinct flipped variables, regardless of cycles in the factor graph.

Cycle encounters are automatic repairs. Suppose the cascade reaches a support clause c for some variable v, and one of the $k-1$ other variables in c, say x_j, was already flipped earlier in the cascade. Since c is a support clause for v, every literal in c other than v's is false under y^*; in particular, x_j appears in c with the sign opposite to y_j^*. Flipping x_j from y_j^* to $\neg y_j^*$ therefore makes x_j's literal in c true, satisfying c without any additional flip.

Stochastic domination. Each such automatic repair removes one branch from the cascade tree T (the sub-cascade that would have followed from flipping a fresh repair variable for c). The cascade on the full graph is therefore a strict pruning of the cascade on the local tree: every cycle encounter removes branches, never adds them. The extinction probability satisfies $q_{\mathrm{graph}} \ge q_{\mathrm{tree}} = q$.

Cycle sparsity. Define the directed cascade graph G_C: variable x_a points to variable x_b if there exists a support clause for x_a containing x_b. Each variable has expected out-degree $e_{\mathrm{sup}}(k-1)$ in G_C. The probability that a specific pair (x_a, x_b) shares a cascade edge is $e_{\mathrm{sup}}(k-1)/n$. A directed cycle of length ℓ through ℓ specified variables occurs with probability $(e_{\mathrm{sup}}(k-1)/n)^\ell$. The expected number of directed ℓ-cycles is $(e_{\mathrm{sup}}(k-1))^\ell/\ell$, and the expected number of such cycles *touching a specific variable* is $O\big((e_{\mathrm{sup}}(k-1))^\ell/(n\ell)\big) = O(1/n)$ for fixed ℓ. Summing over $\ell = 2, \ldots, L$ for any fixed L: the expected number of cascade branches pruned at any given variable is $O(1/n)$.

Supercriticality is preserved. The effective offspring distribution at each variable is $S_v - R_v$, where $S_v \sim \mathrm{Poi}(e_{\mathrm{sup}})$ is the support count and $R_v \ge 0$ is the number of automatically repaired clauses. Since $\mathbb{E}[R_v] = O(1/n)$, the effective offspring mean is $e_{\mathrm{sup}} - O(1/n) > 1$ for n sufficiently large. The branching process remains supercritical, so $q_{\mathrm{graph}} < 1$ and the cascade diverges with positive probability.

Therefore no finite-length path in G_{SAT} can flip x_i while maintaining satisfiability. □

Remark 11 (Cycle robustness)**.** *The proof of (b) establishes cycle robustness via two independent mechanisms. First, the first-passage property ensures the cascade explores a tree of distinct variables even on a graph with cycles. Second, cycle encounters in support clauses produce* deterministic *free repairs (the sign always cooperates), which can only help the adversary—but the help is negligible ($O(1/n)$ branches pruned per variable). For cryptographic applications, KeyGen can additionally verify in polynomial time that the designated frozen coordinate i has a cycle-free cascade neighborhood (only $O(1)$ supported variables fail this check, while $\Theta(fn)$ are available), providing a belt-and-suspenders guarantee. Even without this check, the stochastic domination argument shows $f_{\mathrm{graph}} = f_{\mathrm{tree}} - O(1/n)$, so Theorem 1 holds unconditionally on the full graph.*

Remark 12 (Cascade example)**.** *Consider $k = 3$ with $y^* = (1,1,1,1,1,1)$ and clauses $c_1 = (x_1 \vee \neg x_2 \vee \neg x_3)$, $c_2 = (x_2 \vee \neg x_4 \vee \neg x_5)$, $c_3 = (x_3 \vee \neg x_5 \vee \neg x_6)$. Under y^*, variable x_1 is the unique satisfier of c_1 (the negated literals $\neg x_2, \neg x_3$ are false). Flipping x_1 breaks c_1; repairing c_1 by flipping x_2 (making $\neg x_2$ true) breaks c_2 (where x_2 was the unique satisfier); repairing c_2 by flipping x_4 breaks further clauses. Each repair triggers new breaks—the cascade propagates through the factor graph.*

Remark 13. *For $k = 7$ and $\alpha/\alpha_s = 0.78$: $e_{\sup} = 3.81$, the extinction probability is $q = 0.024$, and the frozen fraction is $f \approx 0.976$. Nearly every variable is frozen.*

3.4 Step 4: Hamming ball containment

Lemma 14. *Every solution $s \in \mathrm{Comp}(y^*)$ satisfies $\mathrm{Ham}(s, y^*) \leq (1 - f + \varepsilon)n$ for any $\varepsilon > 0$, w.h.p.*

Proof. By Lemma 10(b), s agrees with y^* on all $fn - o(n)$ frozen coordinates. Therefore s and y^* differ on at most $(1 - f + \varepsilon)n$ variables. □

3.5 Step 5: Component disjointness

Proof of Theorem 1. Item (i) is Lemma 6. Item (ii) is Lemma 10. Item (iii) is Lemma 14.

For item (iv): suppose $\mathrm{Ham}(y_1, y_2) > 2(1-f)n$. By Lemma 14, $\mathrm{Comp}(y_1) \subset B(y_1, (1-f+\varepsilon)n)$ and $\mathrm{Comp}(y_2) \subset B(y_2, (1 - f + \varepsilon)n)$. By the triangle inequality, $B(y_1, r) \cap B(y_2, r) = \emptyset$ when $2r < \mathrm{Ham}(y_1, y_2)$. Taking $r = (1 - f + \varepsilon)n$ with ε small, the condition $2(1 - f + \varepsilon)n < \mathrm{Ham}(y_1, y_2)$ holds by hypothesis. □

Remark 15 (Generality)**.** *The proof uses only three structural properties: (i) a computable support count exceeding 1, (ii) sparse cycle density in the cascade subgraph (automatic for random hypergraphs at sub-satisfiability density), (iii) disagreeing frozen cores between clusters. Any CSP with these properties—including planted k-XORSAT, planted graph coloring, and planted k-NAE-SAT—satisfies an analogous isolation theorem.*

4 Corollary 2: Quantum Isolation

Proof of Corollary 2. By Theorem 1, G_{SAT} decomposes into connected components $G_1, \ldots, G_m$ respecting cluster boundaries. Define $\mathcal{H}_i = \mathrm{span}\{|s\rangle : s \in G_i\}$. The adjacency matrix A is block-diagonal: $A = \bigoplus_i A_i$. For any analytic g, $g(A) = \bigoplus_i g(A_i)$. This includes e^{-iAt}, any polynomial $p(A)$, and $e^{-A^2 t}$. If $|\psi(0)\rangle \in \mathcal{H}_i$, then $g(A)|\psi(0)\rangle \in \mathcal{H}_i$. □

Remark 16. *This corollary is mathematically elementary; the contribution is the observation that frozen-variable structure in planted CSPs implies quantum isolation, a connection that appears to be new. It rules out any unitary generated by the solution-subgraph structure, not only time evolution. Algorithms operating on the full 2^n-dimensional Hilbert space are addressed in §6.*

5 Commitment Scheme

5.1 Construction

The scheme instantiates a standard sigma protocol [18] with planted k-SAT.

KeyGen. Generate a planted instance Φ with solution y^*. Identify a frozen coordinate i (verifiable by the key generator, who knows y^*, by checking that x_i has ≥ 1 support clause). Public key: (Φ, i). Secret key: y^*.

Commit(b). Produce a non-interactive zero-knowledge proof that "I know y satisfying Φ with $y_i = b$," using the following sigma protocol repeated λ times with Fiat–Shamir [14]:

1. Pick random permutation π on $[n]$ and random sign flip $\sigma \in \{-1, +1\}^n$.
2. Compute $\Phi' = (\pi, \sigma)(\Phi)$ and $y' = (\pi, \sigma)(y^*)$.
3. Send commitment $c = H(\Phi' \| y')$.
4. On challenge $b_c \in \{0, 1\}$: if $b_c = 0$, reveal (π, σ); if $b_c = 1$, reveal y'.

Open(b). Reveal b and the proof transcript.

Verify. Check the proof transcript.

5.2 Security proofs

Proposition 17 (Completeness)**.** *An honest prover succeeds with probability* 1.

Proof. On $b_c = 0$: the prover reveals (π, σ); the verifier recomputes $(\pi, \sigma)(\Phi)$ and confirms it matches the committed Φ'. This always holds by construction. On $b_c = 1$: the prover reveals $y' = (\pi, \sigma)(y^*)$. Since y^* satisfies Φ and (π, σ) preserves satisfiability, y' satisfies Φ'. The verifier confirms this. □

Proposition 18 (Soundness)**.** *A prover who does not know any solution to Φ succeeds with probability at most $2^{-\lambda}$.*

Proof. If the prover can answer both challenges for a single commitment, then from $b_c = 0$ we extract (π, σ) and from $b_c = 1$ we extract y'. Computing $(\pi, \sigma)^{-1}(y')$ yields a solution to Φ—contradicting the assumption. Therefore the prover can answer at most one challenge per round. Over λ independent rounds, the success probability is at most $2^{-\lambda}$. □

Proposition 19 (Zero-knowledge)**.** *There exists a polynomial-time simulator producing transcripts computationally indistinguishable from real transcripts (under collision-resistance of H).*

Proof. The simulator guesses the challenge b'_c uniformly. If $b'_c = 0$: pick random (π, σ), set $\Phi' = (\pi, \sigma)(\Phi)$, commit with a random dummy y_{fake}; if the actual challenge is $b_c = 0$, reveal (π, σ) (correct). If $b'_c = 1$: generate a random planted instance Φ' with a fresh planted solution y', commit $H(\Phi' \| y')$; if $b_c = 1$, reveal y' (correct). On mismatch ($b_c \neq b'_c$), abort and restart. Expected restarts per round: 2.

The simulated transcripts are identically distributed to real transcripts. On $b_c = 0$: both real and simulated reveal a uniform (π, σ). On $b_c = 1$: the real transcript reveals $(\pi, \sigma)(y^*)$, a random solution to a random instance $(\pi, \sigma)(\Phi)$; the simulator reveals a random solution to a random planted instance. These are identically distributed because the planted model is invariant under (π, σ). □

5.3 Binding via frozen-core isolation

Binding. Suppose an adversary opens to both $b = 0$ and $b = 1$. Then the adversary possesses solutions y_0, y_1 to Φ with $(y_0)_i = 0$ and $(y_1)_i = 1$. Since i is frozen, all solutions in $\mathrm{Comp}(y^*)$ satisfy $y_i = y_i^*$ (Definition 4). At least one of y_0, y_1 lies outside $\mathrm{Comp}(y^*)$. Finding such a solution requires solving planted k-SAT independently. Reduction: break binding $\Rightarrow$ solve planted k-SAT.

Remark 20. *Binding and hiding require only that Φ is computationally hard to solve, not that Φ is indistinguishable from random. For applications requiring indistinguishability (CCA security), parameters should be chosen in the quiet planting window [1]; for $k \geq 15$ this window overlaps with the frozen regime.*

Remark 21 (Empirical binding verification)**.** *We verified the binding structure computationally at $n = 20$–75. For each supported variable x_i (i.e., x_i has ≥ 1 support clause), we used Glucose to search for a satisfying assignment with x_i forced to $\neg y_i^*$. The solver succeeds at all sizes tested, but the alternative solutions lie in a* different cluster*: the minimum Hamming distance from y^* satisfies $d_{\min}/n \geq 0.34$ at every size tested, with average $d/n \approx 0.45$ (Table 2). This confirms the binding mechanism: flipping a supported variable within $\mathrm{Comp}(y^*)$ is blocked by the cascade structure; the only route to a solution with $x_i \neq y_i^*$ is to solve planted k-SAT independently and land in a distant cluster. The cost of this independent solve is the classical attack cost $2^{0.234n}$ established in §7.5.*

Table 2: Cluster separation distance under forced variable flip. For each n, 5 supported variables are flipped; the solver's alternative solution is measured against y^*. Separation is $\Omega(n)$: $d_{\min}/n \geq 0.34$ at all sizes.

n	Tests	$d_{\min}$	d_{avg}	$d_{\min}/n$	d_{avg}/n
20	5	8	9.6	0.400	0.480
30	5	16	17.0	0.533	0.567
40	5	18	20.8	0.450	0.520
50	5	17	21.4	0.340	0.428
60	5	26	27.4	0.433	0.457
75	3	28	33.3	0.373	0.444

5.4 Concrete parameter selection

We recommend $k = 7$ (minimum k for strong freezing) at operating density $\alpha/\alpha_s = 0.78$ (deeply supercritical, $e_{\mathrm{sup}} = 3.81$, frozen fraction $f \approx 0.976$). The variable count n is sized to resist both classical cryptanalysis (§7.5) and Grover search (§6.6). Empirical benchmarking of Glucose 4 and MiniSat on our exact distribution yields a classical attack cost scaling as $2^{0.23n}$ operations (§7.5); this is the binding constraint, requiring $n \geq \lambda/0.234$.

Public key size. With seed-based key compression (§5.5), the public key is a 32-byte seed plus a 2-byte frozen coordinate index, totaling **34 bytes** at all security levels. Without compression, each clause stores k variable indices ($\lceil \log_2 n \rceil$ bits each) plus k sign bits, yielding 356–776 KB; see Table 4 for the compression ratios.

Commitment size. One SHA-256 hash = 32 bytes.

Proof size. The sigma protocol runs λ rounds. Each round: one 32-byte commitment hash plus a response (either a 32-byte seed encoding (π, σ) or an n-bit assignment y'). Total: approximately $\lambda \cdot (32 + n/8)$ bytes.

Table 3: Concrete parameters at NIST security levels. Public key uses seed-based compression (§5.5). Parameter n is sized by the classical CDCL attack cost $2^{0.234n}$ (§7.5), which dominates the Grover bound $2^{0.488n}$ at these sizes.

Level	λ	n	k	α/α_s	m (clauses)	PK size	Proof size
NIST-1 (128-bit)	128	547	7	0.78	37,854	34 B	12.5 KB
NIST-3 (192-bit)	192	820	7	0.78	56,747	34 B	25.1 KB
NIST-5 (256-bit)	256	1,094	7	0.78	75,708	34 B	42.0 KB

Comparison. The 34-byte public key is smaller than CRYSTALS-Kyber (1,568 bytes at NIST-1). Proof/signature sizes (12.5–42 KB) are comparable to SPHINCS+ (7.9–49.9 KB). The tradeoff is verification time: reconstructing Φ from the seed costs 0.4–1.4 s (Table 4), compared to microsecond verification for lattice-based schemes. For commitment and signature applications where signing/verification latency is not critical, this overhead is acceptable.

5.5 Seed-based key compression

The naïve public key—the full clause list—is large (160–342 KB). We eliminate this overhead entirely by deriving the instance deterministically from a short seed.

1. Sample a uniform seed $s \leftarrow \{0,1\}^{256}$.
2. Derive the planted solution: $y^* = \text{SHAKE256}(s \,\|\, \texttt{"sol"},\ n \text{ bits})$.
3. Derive clause randomness: $r = \text{SHAKE256}(s \,\|\, \texttt{"cls"},\ \text{sufficient length})$.
4. Generate Φ from r using the standard planted model (variable selection by Fisher–Yates shuffle with bytes from r; literal signs sampled from r conditioned on y^* satisfying each clause).
5. Identify a frozen coordinate i: the first variable with ≥ 1 support clause.
6. Public key: (s, i). Secret key: y^*.

The verifier reconstructs Φ from s by executing the identical deterministic procedure. Both parties obtain the same instance; the clause list is never transmitted.

Proposition 22 (Security of seed-based compression)**.** *If* SHAKE256 *is a secure pseudorandom function, then the seed-based scheme is computationally indistinguishable from the scheme with true randomness.*

Proof. Suppose a distinguisher $\mathcal{D}$ can distinguish a planted instance Φ_{seed} generated from SHAKE256(s) from an instance Φ_{rand} generated with true randomness, with advantage ε. Then $\mathcal{D}$ distinguishes the output of SHAKE256 from a uniform random string of the same length with the same advantage ε, contradicting pseudorandomness. All security properties of the commitment scheme (binding, hiding, soundness, zero-knowledge) transfer from the true-randomness setting to the seeded setting with at most negligible loss. □

This is exactly the technique used by CRYSTALS-Kyber, whose public matrix $\mathbf{A}$ is derived from a 32-byte seed ρ via SHAKE128 [11].

At every security level tested, deterministic reconstruction from the 34-byte public key produces a bit-identical instance, with the empirical average support per variable (3.61–3.86) matching the theoretical prediction $e_{\text{sup}} = 3.81$.

Table 4: Seed-based key compression: empirical validation.

Level	n	Uncompressed PK	Compressed PK	Ratio	Reconstruction
NIST-1	547	364,345 B	34 B	10,716×	379 ms
NIST-3	820	546,190 B	34 B	16,064×	840 ms
NIST-5	1,094	794,934 B	34 B	23,380×	1,424 ms

5.6 Digital signature scheme

The sigma protocol of §5.1 yields a digital signature scheme via the standard Fiat–Shamir transform [14], binding the message into the challenge derivation.

Sign(msg**,** y^***).** For $j = 1, \ldots, \lambda$:

1. Pick random (π_j, σ_j). Compute $\Phi'_j = (\pi_j, \sigma_j)(\Phi)$ and $y'_j = (\pi_j, \sigma_j)(y^*)$.
2. Compute commitment $a_j = H(\Phi'_j \| y'_j)$.

Derive challenge bits $(b_1, \ldots, b_\lambda) = H(a_1 \| \cdots \| a_\lambda \| \text{msg})$. For each j: if $b_j = 0$, set $z_j = (\pi_j, \sigma_j)$; if $b_j = 1$, set $z_j = y'_j$. The signature is $\sigma_{\text{sig}} = (a_1, \ldots, a_\lambda, z_1, \ldots, z_\lambda)$.

Verify(msg**,** σ_{sig}**,** (s, i)**).** Reconstruct Φ from seed s. Recompute $(b_1, \ldots, b_\lambda) = H(a_1 \| \cdots \| a_\lambda \| \text{msg})$. For each j: if $b_j = 0$, check that $(\pi_j, \sigma_j)(\Phi)$ is consistent with a_j; if $b_j = 1$, check that y'_j satisfies Φ'_j and is consistent with a_j.

Proposition 23 (Unforgeability)**.** *The signature scheme is existentially unforgeable under chosen-message attack (EUF-CMA) in the random oracle model, assuming planted k-SAT is hard.*

Proof. By the Fiat–Shamir paradigm applied to a proof of knowledge [14, 18]: any forger producing a valid signature on a new message can be rewound (by reprogramming the random oracle) to extract two accepting transcripts for the same commitments with different challenges. From these, as in Proposition 18, one extracts a solution to Φ. Therefore forging a signature reduces to solving planted k-SAT. □

Signature size. Each round contributes 32 bytes (commitment) plus a response: either a 32-byte seed encoding (π_j, σ_j) or an n-bit string y'_j. Total: approximately $\lambda \cdot (32 + n/8)$ bytes—identical to the proof size in Table 3.

6 Quantum Attack Analysis

We prove exponential lower bounds against five quantum algorithmic families. The sixth (Grover) is treated in detail in §6.6.

6.1 Quantum walks

Corollary 2: zero amplitude transfer between clusters for all $t \geq 0$. Exact, not approximate.

6.2 QAOA at all depths

At constant depth $p = O(1)$, the light-cone has radius $O(k^p)$. Clusters disagree on $\Omega(n)$ variables, so overlap is exponentially small. At superconstant depth, the mixing operator restricted to the solution subspace is a polynomial in A; Corollary 2 applies.

6.3 Quantum annealing

Proposition 24. *The minimum spectral gap of the adiabatic interpolation $H(s) = (1-s)H_0 + sH_P$ satisfies $\Delta_{\min} \leq e^{-\Omega(n)}$.*

Proof. At $s = 1$, satisfying assignments in distinct components are degenerate ground states. The transverse field H_0 lifts this degeneracy via tunneling. To tunnel from $\mathrm{Comp}(y_1)$ to $\mathrm{Comp}(y_2)$, at least $\Omega(fn)$ frozen variables must be simultaneously flipped through a region where each contributes energy cost ≥ 1 (one support clause violated per frozen variable). By the Anderson localization bound [3], the tunneling amplitude is $e^{-\Omega(fn)}$ and $\Delta_{\min} \leq e^{-\Omega(n)}$. The adiabatic runtime is $T \geq e^{\Omega(n)}$. □

Remark 25. *The energy cost per frozen variable (≥ 1) exceeds any constant transverse-field strength Γ, confirming that the AKR localization conditions [3] are satisfied at our parameters.*

6.4 Montanaro DPLL walk

Montanaro [22] achieves quadratic speedup over classical backtracking. Classical DPLL on planted k-SAT at our densities requires $2^{\Omega(n)}$ time [1]. Quantum: $2^{\Omega(n/2)}$—still exponential.

6.5 Low-degree polynomials

Bresler–Huang [6]: degree-D polynomial algorithms cannot distinguish planted from random for $D = o(n)$. This subsumes SOS relaxations, statistical queries, and constant-depth quantum circuits.

6.6 Grover's algorithm

Grover's algorithm treats the solution space as an unstructured database, achieving quadratic speedup: $O(2^{n/2}/\sqrt{|S|})$ queries to an oracle that checks satisfiability. For our commitment scheme, the relevant attack is: given Φ and frozen coordinate i, find any solution y with $y_i \neq y_i^*$. This requires finding a solution in a *different* cluster from y^*.

Effective search space. By Theorem 1(iv), solutions outside $\mathrm{Comp}(y^*)$ lie at Hamming distance $> 2(1-f)n$ from y^*. For $k = 7$, $\alpha/\alpha_s = 0.78$: $f \approx 0.976$, so the adversary must find a solution differing from y^* in at least 95.2% of variables. Each such solution resides in a cluster whose discovery constitutes an independent planted k-SAT instance. Grover provides no structural advantage beyond brute-force search over this effective space.

Query complexity lower bound. Grover's algorithm is optimal for unstructured search [4, 25]. Any quantum algorithm making T queries to a satisfiability oracle succeeds with probability at most $O(T^2|S|/2^n)$. For our parameters, $|S|/2^n$ is exponentially small. Since the adversary must find a solution outside $\mathrm{Comp}(y^*)$, and each cluster is confined to a Hamming ball of radius $(1-f)n$, the effective Grover complexity is $\Omega(2^{fn/2}) = \Omega(2^{0.488n})$.

Parameter sizing. For λ-bit security against Grover: set $n \geq 2\lambda/f$. However, classical CDCL attacks scale as $2^{0.234n}$ (§7.5), which is the binding constraint at all NIST levels. The recommended parameters (Table 3) are sized by the classical bound $n \geq \lambda/0.234$, which automatically satisfies the weaker Grover requirement $n \geq 2\lambda/f \approx 2.05\lambda$. Compare: AES-128 requires doubling key length to 256 bits for Grover resistance; our classical bottleneck absorbs the quantum threat with margin.

Oracle cost. Each Grover oracle call requires evaluating Φ on a candidate assignment: checking $m = \alpha n$ clauses of width k. This costs $O(k\alpha n) = O(n)$ classical gates per oracle call, or $O(n)$ Toffoli gates in a reversible circuit. The total quantum circuit depth is $O(n \cdot 2^{fn/2})$, which is exponential. The constant factors from reversible arithmetic are comparable to those in Grover-on-AES analyses [16].

Standard treatment. Every deployed and proposed post-quantum scheme handles Grover identically: parameter doubling. CRYSTALS-Kyber, CRYSTALS-Dilithium, SPHINCS+, McEliece—all set parameters to absorb the $\sqrt{\cdot}$ speedup. No scheme has a structural Grover barrier because no such barrier exists for any NP search problem [4]. Our treatment is therefore exactly on par with the NIST-standardized schemes. The critical distinction is that the *other* five quantum attack families (Table 5) *do* admit structural barriers, which we prove. Grover is the only residual threat precisely because it ignores all problem structure.

6.7 Summary

Table 5: Quantum attack analysis. Five of six major quantum algorithmic families face proved structural barriers. Grover's algorithm, the sole remaining threat, is mitigated by standard parameter doubling (§6.6), identically to how all NIST-standardized PQC schemes handle unstructured quantum search.

Attack	Model	Barrier	Status
Quantum walks	G_{SAT}	Cor. 2	Proved
QAOA (all depths)	G_{SAT}	Cor. 2 + light cone	Proved
Quantum annealing	2^n	$\Delta \leq e^{-\Omega(n)}$	Proved
Montanaro DPLL	DPLL tree	$2^{\Omega(n/2)}$	Proved
Low-degree poly	Statistical	[6]	Proved
Grover	2^n	$\Omega(2^{0.488n})$	Mitigated
Arbitrary BQP	2^n	Assumption	Same as all PQ

We emphasize that no existing post-quantum scheme—including CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+—provides a structural Grover barrier. Our scheme matches this standard while providing proved barriers against five additional quantum attack families that lattice-based and hash-based schemes do not explicitly address.

7 Empirical Validation

7.1 Multi-seed validation ($n = 16$)

We generated planted instances with two solutions y_1, y_2 at Hamming distance $n/2$ across 100 random seeds at $n = 16$, $k = 7$, $\alpha/\alpha_s = 0.78$. For each seed we enumerated all satisfying assignments, clustered by proximity, built the Hamming-1 solution subgraph, and checked for shared connected components between clusters.

At every seed tested, the minimum inter-cluster Hamming distance equals n and no connected component contains solutions from both clusters.

Table 6: Multi-seed validation: $n = 16$, $k = 7$, $\alpha/\alpha_s = 0.78$, 100 seeds.

Metric	Value
Valid instances (both clusters non-empty)	100/100
Min inter-cluster distance (every seed)	16 ($= n$)
Seeds with shared C_1/C_2 components	0/100
Complete cluster isolation	100/100

7.2 Scaling ($n = 14$–22, single seed)

Table 7: Barrier scaling at $k = 7$, $\alpha/\alpha_s = 0.78$, seed 42.

n	$\|S\|$	$\|C_1\|$	$\|C_2\|$	MinInter	Comp	Peak $\|C_2\|_2$
14	12	6	6	14	11	0
16	13	7	6	16	11	0
18	20	12	8	18	19	0
20	27	11	16	20	25	0
22	39	24	15	22	33	0

7.3 Phase transition in k

Table 8: Phase transition in k: $n = 20$, $\alpha/\alpha_s = 0.65$, seed 42. Lower density than Table 7 to show that barrier onset depends primarily on k. The $k = 9$ entry has MinInter $= 2$ yet zero leakage, confirming that the mechanism is component fragmentation.

k	$\|S\|$	MinInter	Comp	Peak $\|C_2\|_2$
5	82	3	25	4.7×10^{-1} (Leaks)
7	27	20	25	0 (Blocked)
9	140	2	116	0 (Blocked)

7.4 Hop stress test

7.5 Concrete classical cryptanalysis

To establish concrete bit-security, we benchmarked two state-of-the-art CDCL solvers—Glucose 4 and MiniSat 2.2—on planted instances at our exact operating parameters ($k = 7$, $\alpha/\alpha_s = 0.78$), averaging over 5 random seeds per size. Instances were generated using the seed-based procedure of §5.5.

Exponential fit. Fitting $\log_2(\text{time}) = a + \beta n$ on the non-timeout data ($n = 30$–60) gives:

- Glucose: $\beta = 0.234$, $R^2 = 0.985$.
- MiniSat: $\beta = 0.237$, $R^2 = 0.956$.

Table 9: Hop sweep: $n = 22$, $k = 7$, $\alpha/\alpha_s = 0.78$. Barrier holds through hop 4.

Hop	Edges	Comp	Peak $\lvert C_2\rvert_2$	
1	15	30	0	Blocked
2	27	29	0	Blocked
4	27	29	0	Blocked
5	66	12	3.4×10^{-2}	Leaks

Table 10: CDCL solver performance on planted k-SAT at $k = 7$, $\alpha/\alpha_s = 0.78$. Average over 5 seeds. Timeout: 30 s.

n	Clauses	Glucose time	Glucose conflicts	MiniSat time	MiniSat conflicts
30	2,076	0.024 s	2,559	0.007 s	1,103
40	2,768	0.070 s	6,679	0.031 s	4,245
50	3,460	0.610 s	32,678	0.381 s	37,973
60	4,152	2.628 s	124,874	0.698 s	61,192
75	5,190	> 30 s	975,780	> 30 s	733,584

Fitting on conflict counts (a hardware-independent metric):

- Glucose conflicts: $\beta = 0.191$, $R^2 = 0.992$.
- MiniSat conflicts: $\beta = 0.205$, $R^2 = 0.952$.

Taking the most conservative (attacker-favorable) fit, $\beta = 0.234$, the classical attack cost is $2^{0.234n}$ operations.

Table 11: Extrapolated classical bit-security at recommended parameters. Growth rate $2^{0.234n}$ from Glucose CDCL fit.

Level	n	Classical cost	Grover cost	Security target
NIST-1	547	2^{128}	2^{267}	2^{128}
NIST-3	820	2^{192}	2^{400}	2^{192}
NIST-5	1,094	2^{256}	2^{534}	2^{256}

Extrapolated bit-security. At the recommended parameters, the classical CDCL attack cost meets the NIST security target at each level, and the Grover (quantum brute-force) cost exceeds it by a wide margin. The classical bottleneck dominates: parameters are sized by the $2^{0.234n}$ classical bound, not the $2^{0.488n}$ quantum bound.

Remark 26. *The growth rate $\beta \approx 0.23$ is consistent with known hardness results for planted k-SAT in the frozen regime. CDCL solvers are the strongest known general-purpose classical attack; Survey Propagation and Belief Propagation guided decimation are known to fail above the freezing threshold [1]. We are not aware of any classical algorithm achieving $\beta < 0.20$ on our distribution.*

8 Discussion

8.1 Relationship to the Overlap Gap Property

Theorem 1 is related to but distinct from Gamarnik's Overlap Gap Property (OGP) [15]. The OGP shows that solution spaces lack smooth interpolation paths between distant configurations. Theorem 1 is strictly stronger in one sense: it proves topological disconnection, not merely absence of smooth paths. However, the OGP applies to a broader class of problems. Our result can be viewed as a rigorous strengthening of the OGP for planted k-SAT above the planted freezing threshold.

8.2 A fourth hardness family

Lattices, hash functions, codes, and now frozen-core constraint satisfaction. The structural evidence for quantum resistance (Table 5) is, to our knowledge, more explicit in its coverage of specific quantum algorithmic families than what is available for any existing post-quantum scheme. The recent proposal of quantum stabilizer decoding as a PQC hardness family [19] and the present work both suggest that the PQC assumption space is richer than the current NIST portfolio of lattices, hashes, and codes. CSP cluster structure and quantum code decoding represent two independent directions for diversification.

8.3 On the hardness assumption

Our binding security reduces to the average-case hardness of planted k-SAT at specific parameters. We acknowledge that a worst-case to average-case reduction—analogous to Regev's reduction from GapSVP to LWE—would strengthen the proposal, and we list this as an open problem. We note, however, that this limitation is shared by multiple deployed and standardized PQC schemes:

- Hash-based signatures (SPHINCS+/FIPS 205) rest on the assumption that SHA-256 is collision-resistant. No worst-case reduction to any NP-hard problem is known.
- Code-based encryption (Classic McEliece) rests on the average-case hardness of decoding random linear codes.
- Lattice-based schemes, while enjoying worst-case reductions in principle, derive their concrete parameter security from heuristic cost models of the BKZ algorithm at specific block sizes—an average-case extrapolation comparable to our CDCL benchmarks.

Our concrete security analysis (§7.5) follows the same methodology used in the lattice PQC literature: benchmark the best known attack at small parameters, fit an exponential, and extrapolate. The $R^2 = 0.985$ fit quality is comparable to or better than typical BKZ cost model fits.

8.4 Generality

By Remark 15, the isolation theorem applies to any CSP with frozen-variable structure. If planted k-SAT is broken by a future SAT-specific algorithm, the framework survives with a different frozen-core CSP.

8.5 Information-geometric connection

The topological fragmentation proved here corresponds to a divergence of the Fisher–Rao metric on the Bernoulli manifold at the cluster boundary. This connection will be developed in forthcoming work.

8.6 Open problems

(i) Prove a full BQP lower bound for planted k-SAT. No existing post-quantum scheme has such a proof; achieving one here would be a first.

(ii) Establish formal relationships (reductions or separations) between the planted k-SAT hardness assumption and existing PQC assumptions (LWE, SIS, syndrome decoding).

(iii) Extend the constructions to a full key encapsulation mechanism (KEM). This requires a lossy mode for planted k-SAT instances, which connects to the quiet planting problem; for $k \geq 15$ this appears feasible.

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